



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## API MANAGEMENT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

**Q1** What is API Management and what is its primary purpose in its cloud ecosystem? **Threat Model**

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**Q6** How do you manage configuration and secrets for API Management? **Observability**

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**Q2** How does IAM work with API Management? **Architecture**

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**Q7** What are the main cost drivers for API Management and how do you optimize them? **Lifecycle**

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**Q3** How do you monitor API Management in production? **Configuration**

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**Q8** How do you implement high availability with API Management? **Compliance**

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**Q4** How do you scale API Management to handle variable load? **Hardening**

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**Q9** How does API Management integrate with a CI/CD pipeline? **Testing**

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**Q5** How do you secure API Management in production? **SSO / IdP**

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**Q10** What are common pitfalls when first adopting API Management? **Tuning**

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 API Management is a managed cloud

Mitigation

API Management is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A6 Store sensitive values in a managed

Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A2 Access to API Management is controlled

Primitives

Access to API Management is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A7 Cost in API Management typically comes

Automation

Cost in API Management typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A3 API Management emits metrics to the

Hardening

API Management emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A8 Deploy API Management across multiple

Governance

Deploy API Management across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A4 API Management supports auto-scaling policies

Gotchas

API Management supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A9 CI/CD pipelines use the cloud CLI

Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to API Management after tests pass. Infrastructure-as-code (Terraform, CDK) defines API Management configuration as version-controlled code deployed via the pipeline.

A5 Place API Management in a private

Federation

Place API Management in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A10 Common mistakes include misconfigured IAM

Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.